Obblivignes KanchanadeviVenkataraman

Ph.D. Student in Computer Science, North Carolina State University
Generative Intelligent Computing (GIC) Lab and Molecular Analytics & Photonics (MAP) Lab
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Research Interests

I build physics-informed generative AI that is judged by how faithfully it models the physical world, starting with spectroscopic single-molecule localization microscopy (sSMLM).

Mainstream AI evaluates diffusion models and U-Nets on benchmark scores such as F1 or FID, which say little about whether an output obeys physics; physical and biomedical imaging, in turn, have been slow to adopt modern generative AI. My work sits in that gap. I pair Monte Carlo simulators and digital twins of real optical systems with denoising and diffusion models, then measure success in physical terms: localization precision, spectral accuracy, robustness at low signal-to-noise, and transfer from simulation to experimental data. In teaching, I hope to bring applied machine learning to students through project-based courses built on real scientific data.

Education

Ph.D., Computer Science
North Carolina State University, Raleigh, NC — Aug 2025 – expected May 2029
- Dissertation (tentative): "Physics-Informed Generative Models and Digital Twins for Efficient Spectroscopic Single-Molecule Imaging"
- Co-advisors: Dr. Dongkuan (DK) Xu, Computer Science (GIC Lab); Dr. Yang Zhang, Textile Engineering, Chemistry & Science (MAP Lab)
- GPA 3.98/4.00

M.S., Computer Science
North Carolina State University, Raleigh, NC — Aug 2023 – May 2025
- Non-thesis · GPA 3.97/4.00

B.S., Computer Engineering
North Carolina State University, Raleigh, NC — Aug 2020 – May 2023
- Summa Cum Laude · University Honors Program Scholar · Dean's List (2021 – 2023) · GPA 4.00/4.00

A.S., Science
Catawba Valley Community College, Hickory, NC — Aug 2016 – May 2020
- Earned concurrently with high school diploma through Challenger Early College High School · High Honors

Teaching Experience

Teaching Assistantships

Graduate Teaching Assistant, CSC 422: Automated Learning and Data Analysis
Department of Computer Science, NC State University — Jan 2026 – present
- Instructor of record: Dr. Xiaorui Liu · ~60 undergraduate students
- Hold 2 office hours per week guiding students through ML concepts, assignments and a semester-long real-world ML project.
- Grade all assignments, quizzes and exams, including designing rubrics and answer keys and adjusting class materials as needed.

Graduate Teaching Assistant, CSC 326: Software Engineering
Department of Computer Science, NC State University — Aug 2025 – Dec 2025

- Instructor of record: Dr. Sarah Heckman · ~60 undergraduate students across two lab sections
- Led two lab sections where student teams built a full-stack application; mentored teams on design, Git workflow and debugging.
- Held office hours on course material and project issues.

Guest Lectures

"Applying a Conditional Diffusion Model Towards Denoising Spectral Imaging"
CSC 791/591: Advanced Topics in Efficient Deep Learning (instructor: Dr. Dongkuan Xu), NC State University — Apr 2026

- Presented research progress on diffusion-based denoising as a case study in efficient deep learning.

Outreach Teaching

Coding Camp Intern
Catawba Valley Community College, Hickory, NC — Jun 2019 – Jul 2019

- Taught elementary and middle school students to program and built insect-bots with them to show how robots are designed and used.
- Developed the camp's interactive software activities and supported students with hardware and software issues.

Research Experience

Graduate Research Assistant
GIC Lab (Dr. Dongkuan Xu, Computer Science) and MAP Lab (Dr. Yang Zhang, Textile Engineering, Chemistry & Science), NC State University — Sep 2024 – present

A connected line of work on physics-based simulation and deep learning denoisers for spectroscopic single-molecule localization microscopy (sSMLM), each project building on the last.

DWP-sSMLM (tentative): dual-wedge-prism digital twin for single-channel spectral imaging — Aug 2026 – present

- Investigating whether the non-linear, wavelength-dependent dispersion of a dual wedge prism (DWP) lets one dispersed image recover both molecular position and emission spectrum, removing the photon-splitting spatial channel of conventional sSMLM.
- Building a Monte Carlo forward model with calibrated DWP dispersion and a wavelength-dependent PSF; validating identifiability with joint maximum-likelihood estimation, Cramér–Rao bounds and ambiguity maps against a linear-disperser control.
- Developing a full-frame DWP simulation pipeline (crop, classify, denoise) that models experimental sSMLM data more faithfully than grating-based simulators. Exploratory, early-stage project.

FastSpecDiffuse (FSD): warm-start initialization for accelerated diffusion inference — Jul 2026 – present

- Initializes SpecDiffuse sampling from an SA_SpecUNet (or alternative) estimate instead of pure noise, outperforming naive DDPM timestep respacing.
- Cut sampling from 5–10 to 2–3 timesteps with no loss in denoising performance on simulated and experimental data.
- Ongoing: comparing initializers and characterizing the mechanism by which warm starts improve inference.

SpecDiffuse: robust conditional denoising diffusion model for sSMLM — Jan 2025 – present

- Built a lightweight conditional diffusion (DDPM) model on an SA_SpecUNet backbone, extending the published EMDiffuse framework to sSMLM; built from scratch in Python/PyTorch.
- Trained on DWP-based physics simulations; transfers to experimental data and performs especially well on low-SNR datasets.
- Improved resolution ~50% over prior U-Net implementations on low-SNR datasets.

SA_SpecUNet: spatial-attention residual U-Net with multi-scale background simulation — Jan 2025 – present

- Built a multi-scale Perlin-noise background simulator for more realistic training data.
- Added spatial attention gates inspired by Attention U-Net (Oktay et al., 2018) and reframed the task as residual prediction rather than direct background prediction.
- Faster and more accurate than comparison methods, improving resolution ~25% over SpecUNet.

SpecUNet: supervised deep learning framework for single-molecule spectroscopic image denoising — Sep 2024 – Jul 2025

- Developed grating-based Monte Carlo simulations (100k+ images) to generate ground truth for sSMLM data.
- Trained a U-Net (built in MATLAB, then ported to Python/PyTorch for configurability and scalability) to predict background and denoise spectral images, improving resolution ~70% over conventional methods.
- Published in *Analytical Chemistry* (2025, cover article) and *Proc. SPIE* (2026).

SIMBA: agentic AI platform for single-molecule multidimensional imaging — Aug 2025 – present · project website

- Led the denoising module: packaged the lab's denoisers as Python packages and built their integration into the SIMBA application.
- Integrated the published SpecUNet; SA_SpecUNet, SpecDiffuse and FastSpecDiffuse are planned next.
- SIMBA is a multi-agent desktop application (Electron, React, GPT-4o, LiteLLM) whose LLM agents interpret user intent and orchestrate localization, spectral analysis and denoising tools.

Graduate Research Assistant, Software Engineering AI

Dr. Bowen Xu's Lab, NC State University — Jan 2024 – May 2024

- Trained and evaluated a deep just-in-time code-comment consistency checker on datasets including Microsoft CodeXGLUE.
- Integrated the tool into the SEEDGuard repository with video tutorials and Docker images.

Undergraduate Research Assistant, Distributed Systems

Dr. Ruozhou Yu's Lab, NC State University — Jan 2023 – Aug 2023

- Benchmarked microservice applications on a Raspberry Pi Kubernetes cluster (Cassandra, Flask, Jaeger, OpenTelemetry); project repository.
- Trained ML models predicting service performance, reaching $R^2 = 0.96$ with random forests.

Publications

Peer-Reviewed Journal Articles and Conference Proceedings

1. Mao, H., **KanchanadeviVenkataraman, O.**, Xu, D., Song, K.-H., & Zhang, Y. (2026). A Monte Carlo-based deep learning framework for accurate single-molecule spectroscopic imaging. *Proceedings of SPIE*, *Single Molecule Spectroscopy and Superresolution Imaging XIX*, 13858. doi:10.1117/12.3082055
2. Mao, H., Liu, Y., **KanchanadeviVenkataraman, O.**, Shahid, M. A., Laplante, C., Xu, D., Song, K.-H., & Zhang, Y. (2025). Framework for accurate single-molecule spectroscopic imaging analyses using Monte-Carlo simulation and deep learning. *Analytical Chemistry*, 97(30), 16250–16258. Cover article. doi:10.1021/acs.analchem.5c01486

Preprints

1. Mao, H., Mauny, H., **KanchanadeviVenkataraman, O.**, Laplante, C., Xu, D., & Zhang, Y. (2026). SIMBA: An agentic AI platform for single-molecule multidimensional imaging. *bioRxiv*, 2026.04.16.719005. doi:10.64898/2026.04.16.719005

Manuscripts in Preparation

1. **KanchanadeviVenkataraman, O.**, Mao, H., Xu, D., & Zhang, Y. FastSpecDiffuse: Warm-start initialization for few-step diffusion denoising in spectroscopic single-molecule localization microscopy (working title).
2. Mao, H.∗, **KanchanadeviVenkataraman, O.∗**, Xu, D., & Zhang, Y. SpecDiffuse: A robust conditional denoising diffusion model for spectroscopic single-molecule localization microscopy. (∗equal contribution)

3. Mao, H., **KanchanadeviVenkataraman, O.**, Xu, D., & Zhang, Y. SA_SpecUNet: Spatial-attention residual learning for fast and accurate spectroscopic single-molecule image denoising (working title).

Conferences, Presentations & Posters

Posters

1. **KanchanadeviVenkataraman, O.**, Mao, H., Xu, D., & Zhang, Y. "SpecDiffuse: A Robust Conditional Denoising Diffusion Model for Spectroscopic Single-Molecule Localization Microscopy." *2026 Applied AI Symposium*, NC State College of Engineering, Raleigh, NC, Sep 23, 2026.

Invited and External Talks

1. "SIMBA: An Agentic AI Platform for Single-Molecule Multidimensional Imaging." Invited virtual presentation, The University of Texas MD Anderson Cancer Center, Dec 19, 2025.
2. "Deep Learning Denoisers for Spectroscopic SMLM: SpecUNet, SA_SpecUNet and EMDiffuse." *2nd Annual MARINA Workshop* (Marine Artificial Intelligence and Research Alliance), NC State University, Raleigh, NC, Aug 15, 2025.

Research Group Presentations

1. "Physics-Informed Diffusion Models." GIC Lab research seminar, Department of Computer Science, NC State University, Aug 2026.
2. Recurring research-progress presentations, GIC Lab and MAP Lab, NC State University, 2024 – present.

Grants, Honors & Awards

Research Funding Supporting My Work (contributing researcher)

- 2026 – 2027 — NC State College of Engineering Dean's Applied AI Research Accelerator Award, "Transforming Nanoscale Sensing with Scalable Agentic AI and Multi-Functional Microscopy" (PI: D. Xu; Co-PIs: Y. Zhang, C. Laplante) — $100,000
- 2025 – 2026 — NC State Comparative Medicine Institute (CMI) Ideation Award, "Developing an Agentic AI Framework for Single-Molecule Super-Resolution Imaging Analytics" (Co-PI: D. Xu) — $30,000

Honors and Awards

- Aug 2025 — Cover article, *Analytical Chemistry*, Vol. 97, Issue 30
- May 2023 — Summa Cum Laude, B.S. Computer Engineering, NC State University
- 2021 – 2023 — Dean's List, NC State University
- 2020 – 2023 — University Honors Program Scholar, NC State University
- May 2020 — High Honors, A.S. Science, Catawba Valley Community College
- Feb 2020 — 4th Place, North Carolina Mathematical Association of Two-Year Colleges (NCMATYC) Calculus Tournament
- Mar 2019 — 1st Place, Advanced Math Division, Jeanette L. Staley Math Tournament
- Mar 2019 — 3rd Place, Water Quality Event, Science Olympiad (District Level)
- Mar 2018 — 3rd Place, Math III Division, Jeanette L. Staley Math Tournament
- Mar 2017 — 4th Place, Math II Division, Jeanette L. Staley Math Tournament

Service

- 2026 — Reviewer (3 papers), International Workshop on Resource-efficient Learning for Knowledge Discovery (RelKD 2026), held with ACM SIGKDD 2026, Jeju, Korea
- Aug 2021 – Dec 2022 — Electrical and Computer Engineering Student Ambassador, NC State University: led campus tours for prospective students, organized student–faculty events, and tutored ECE students

- Jul – Aug 2019; Jun – Aug 2016 — Volunteer, Catawba County Museum of History, Hickory, NC (50 hours): analyzed 3,000+ visitor records to profile the museum's national, state and local reach (49 states, 18 countries) and ran a matching volunteer analysis
- Mar 2019 — Volunteer, Hickory Soup Kitchen, Hickory, NC (15 hours): prepared and served meals and stocked food donations

Professional Development

- Expected Oct 2026 — Teaching Certificate, Level 1 (Basic), NC State Graduate School
  - Foundational workshops (2026): Teaching College (Jan 16); Using Learning Outcomes (Jan 23); Engaging Diverse Learners (Feb 20); Mentoring 101 (Mar 13)
  - Electives (2026): Teaching in an AI Environment (Feb 27); Processing Course Evaluations (Mar 27); Writing a Teaching Philosophy (Apr 3); Microlearning > Teaching Moodle course (Jul)
- Sep 2026 — 2026 Applied AI Symposium (presenter), NC State College of Engineering
- Jul 2026 — Course Design Essentials, NC State University
- Oct 2025 — CITI Responsible Conduct of Research, CITI Program (valid through Oct 2028)
- Aug 2024 — IBM Data Science Professional Certificate, IBM (Coursera)
- Aug 2023 — AWS Knowledge: Cloud Essentials, Amazon Web Services
- Aug 2022 — AWS Machine Learning Foundations Nanodegree, Udacity

Professional Affiliations

- 2018 – present — Member, Phi Theta Kappa Honor Society (Catawba Valley Community College chapter)

Skills and Languages

- Spoken languages: English (full professional), French (professional working), Tamil (limited working), Kannada (limited working)
- Programming: Python, MATLAB, C, C++, Java, JavaScript, TypeScript, LaTeX, Bash
- Frameworks and tools: PyTorch, Keras, scikit-learn, NumPy, SciPy, Pandas, Matplotlib, Plotly, React, Electron, Git, SLURM
- Methods: diffusion models (DDPM), attention U-Nets, LLM agents (GPT-4o, LiteLLM), Monte Carlo simulation, maximum-likelihood estimation, statistical analysis

References

- **Dr. Dongkuan (DK) Xu** — Assistant Professor, Department of Computer Science, NC State University (Ph.D. co-advisor) — dxu27@ncsu.edu — i.dk-lab.xyz
- **Dr. Yang Zhang** — Assistant Professor, Department of Textile Engineering, Chemistry & Science, NC State University (Ph.D. co-advisor) — yzhan296@ncsu.edu — faculty page
- **Dr. Xiaorui Liu** — Assistant Professor, Department of Computer Science, NC State University (instructor of record, CSC 422) — xliu96@ncsu.edu — faculty page